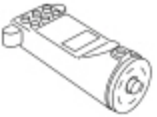


Auxiliary Brake System — Trailer Brake Control (TBC) Module

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2954-A	7-Way Trailer Emulator TEK-6562A or equivalent

Principles of Operation

NOTE: A trailer brake emulator tool is available to verify that the Trailer Brake Control (TBC) module and vehicle wiring are functioning correctly. The trailer emulator requires the trailer tow connector ground and 12 volt trailer tow charge circuit to be connected in order to check the electric brake circuit. Battery Junction Box (BJB) fuse 21 (30A) and the Trailer Tow/Battery Charge Relay must be installed for battery voltage to be supplied to the trailer tow connector. The fuse and relay are supplied as loose-shipped items with the vehicle and may not have been installed during Pre-Delivery Inspection (PDI) .

The Trailer Brake Control (TBC) module is an integrated electronic control device designed to provide variable braking power to the electric-actuated drum brakes on a towed trailer (1 to 4 axles only). The braking energy provided to the trailer is varied with a Pulse Width Modulated (PWM) signal that switches between 0 volts and battery voltage, the higher the duty cycle the more braking power available.

The TBC module receives information from the stoplamp switch and the PCM. The brake pressure transducer is internal to the ABS module.

The stoplamp switch is mounted on the brake pedal bracket and switches on or off depending on the brake pedal position. The TBC module receives the stoplamp switch and the brake torque information over the High Speed Controller Area Network (HS-CAN) from the PCM.

The **TBC** module receives information through the **HS-CAN**. These inputs are from the PCM, the Instrument Panel Cluster (IPC) and the ABS module. The Body Control Module (BCM) also sends ignition state, park lamp status and illumination dimming level data to the **TBC** module. The information sent from the **TBC** module is passed to the **IPC** and is displayed on the message center.

To adjust the gain setting up or down, the **TBC** module faceplate provides the driver with control buttons which determine the maximum power output. The **TBC** module faceplate also provides a manual control lever to manually apply the trailer brakes, independent of the vehicle brakes, to calibrate the gain setting to the specific trailer loading and road conditions. When the manual control lever is activated with a trailer connected to the vehicle, the trailer stoplamps and the vehicle stoplamps (except the center high mount stoplamp) illuminate. If a trailer is not connected and the manual control lever is activated, the vehicle still illuminates the vehicle stoplamps (except the center high mount stoplamp).

Trailer Brake Control (TBC) Module Wake Up

NOTE: *These conditions require that a trailer **not** be connected to the trailer tow connector.*

The following conditions describe the normal operation of the **TBC** module at start up.

When the ignition key is in the RUN position and the manual lever is set all the way to the left with a gain of 10, the message center should display TRAILER DISCONNECTED. At the same time, with the vehicle stationary, battery voltage should be present at pin 3 of the trailer tow connector.

When not braking the **TBC** module sends a voltage pulse every 4 seconds to pin 3 of the trailer tow connector to determine whether a trailer is connected to the trailer tow connector.

Instrument Panel Cluster (IPC) Configuration For Trailer Brake Control (TBC) Functionality

The same **IPC** is used for vehicles with and without the **TBC** option. **TBC** functionality is enabled or disabled using the scan tool. If the vehicle is equipped with a **TBC** module, but the **IPC** was not configured for **TBC**, the **TBC** module shuts down, no **TBC** information is displayed in the message center and DTC U2202 sets in the **TBC** module.

If the **IPC** is configured for **TBC**, but the vehicle is not equipped with a **TBC** module, TRAILER BRAKE MODULE FAULT is displayed in the message center. This symptom could occur if a replacement **IPC** was installed, but not configured correctly. For information on the **TBC** programmable parameter, refer to Programmable Parameters below.

Programmable Parameters

The message center displays **TBC FAULT** if it is receiving communications from the **TBC** module, but has not been configured for a trailer brake system. If the message center displays **TBC FAULT** and there are no DTCs stored in the **TBC** module, follow these steps to verify and/or modify the **TBC** module programmable parameter in the Instrument Panel Cluster (IPC).

1. Identify the vehicle using the normal Integrated Diagnostic System (IDS) process.
2. Select Programming and press the check box.
3. Select Module Programming and press the check box.
4. Select Programmable Parameters and press the check box.
5. Select Personality and press the check box.
6. Follow the on-screen directions.
7. **NOTE:** *The scan tool displays either EQUIPPED or NOT EQUIPPED to the left of **TBC -IPC**.*

Highlight **TBC -IPC**.

8. If the **IPC** needs to be configured for a trailer brake system, choose EQUIPPED and press the check box.

9. Clear DTCs from the TBC module, then carry out a self-test on the TBC module. Refer to the Trailer Brake Control (TBC) Module DTC Chart for any DTCs retrieved.

Inspection and Verification

NOTE: Carry out all pinpoint tests *without* a trailer connected to the vehicle.

NOTE: A trailer brake emulator tool is available to verify that the TBC module and vehicle wiring are functioning correctly. The trailer emulator requires the trailer tow connector ground and 12 volt trailer tow charge circuit to be connected in order to check the electric brake circuit.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Base brake system • Trailer	• Battery Junction Box (BJB) fuse 17 (30A) • Body Control Module (BCM) fuse 37 (10A) • Wiring, terminals or connectors • Trailer Brake Control (TBC) module

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** Make sure to use the latest scan tool software release.

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
- verify the ignition is ON.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.
- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record Continuous Memory Diagnostic Trouble Codes (CMDTCs) .

8. Clear the CMDTCs and carry out the self-test diagnostics for the TBC module.

9. If the DTCs retrieved are related to the concern, go to the Trailer Brake Control (TBC) Module DTC Chart. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Chart

Trailer Brake Control (TBC) Module DTC Chart

DTC	Description	Action
B1317	Battery Voltage High	GO to Pinpoint Test D.
B1318	Battery Voltage Low	GO to Pinpoint Test E.
B1342	ECU is Faulted	INSPECT for a stuck or damaged manual sliding lever and gain control buttons. REPAIR or CLEAR any obstructions or foreign material. REPAIR all other DTCs before repairing B1342. CLEAR the DTCs. TEST the system for normal operation. NOTE: <i>It takes 2 minutes for the Trailer Brake Control (TBC) module to set this DTC due to a stuck manual sliding lever or a stuck gain control button.</i> If the DTC returns, INSTALL a new Trailer Brake Control (TBC) module. REFER to Trailer Brake Control (TBC) Module in this section.
B1863	TBC Ground Open Circuit	GO to Pinpoint Test F.
C2806	Trailer Brakes Wiring Circuit Short to Vbatt	GO to Pinpoint Test A.
C2807	Trailer Brakes Wiring Circuit Short to Ground	GO to Pinpoint Test A.
U0100	Lost Communication With ECM /PCM "B"	GO to Pinpoint Test G.
U0121	Lost Communication With Anti-Lock Brake System (ABS) Control Module	GO to Pinpoint Test H.
U0140	Lost Communication With Body Control Module (GEM)	GO to Pinpoint Test I.
U0155	Lost Communication With Instrument Panel Cluster (IC) Control Module	GO to Pinpoint Test J.
U0415	Invalid Data Received from Anti-Lock Brake System (ABS) Control Module	This DTC indicates that the TBC module has received the brake pressure information sent from the ABS module, but that information does not agree with other sensor inputs. REFER to Section 206-09 to diagnose the ABS module.
U2202	Invalid Configuration Data Received	The Instrument Panel Cluster (IPC) is not configured for trailer brakes. REFER to Programmable Parameters in this section. CLEAR the DTCs. REPEAT the self-test.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action

No communication with the Trailer Brake Control (TBC) module	- Fuse(s) - Wiring, terminals or connectors - TBC module	REFER to Section 418-00.
The trailer brakes are inoperative	- Fuse(s) - Wiring, terminals or connectors - Trailer tow connector - TBC module - Customer trailer	GO to Pinpoint Test B.
The trailer brakes are always engaged	- Wiring, terminals or connectors - TBC module - Customer trailer	GO to Pinpoint Test C.
The message center intermittently displays TRAILER DISCONNECTED or WIRING FAULT ON TRAILER while towing a trailer	- Wiring, terminals or connectors - Customer trailer	GO to Pinpoint Test A.
The message center displays TBC FAULT	DTCs present	RETRIEVE and RECORD the DTCs. REFER to the Trailer Brake Control (TBC) Module DTC Chart.
The message center displays TRAILER BRAKE MODULE FAULT when the vehicle is not equipped with a TBC module	Incorrect IPC configuration	CONFIGURE the IPC . REFER to Programmable Parameters in this section.
The message center displays WIRING FAULT ON TRAILER	- Wiring, terminals or connectors - TBC module - Customer trailer	GO to Pinpoint Test A.

Pinpoint Tests

Pinpoint Test A: The Message Center Displays WIRING FAULT ON TRAILER

NOTE: Carry out all pinpoint tests *without* a trailer connected to the vehicle.

Refer to Wiring Diagrams Cell [95](#) , Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The braking energy provided to the trailer is varied with a Pulse Width Modulated (PWM) signal that switches between 0 volts and battery voltage, the higher the duty cycle the more braking power available. The Trailer Brake Control (TBC) module varies the PWM signal based on various inputs such as the brake torque message, the manual slider switch and the gain buttons. With the vehicle stationary and the manual slider switch fully to the left with a gain of 10, there should be more than 10-12 volts supplied to pin 3 of the 7-pin trailer tow connector. The TBC module Pulse Width Modulated (PWM) output voltage is speed dependent. With the vehicle stationary and the brake pedal applied, there is approximately 0 volts at the trailer tow connector (pin-3 or pin-7). When the trailer is connected to the vehicle the Instrument Panel Cluster (IPC) should show the TRAILER CONNECTED message. If the WIRING FAULT ON TRAILER is only message displayed when the trailer is connected, the fault is with the trailer wiring.

Every 4 seconds (when not braking) the TBC module sends a voltage pulse to pin 3 of the 7-pin trailer tow connector to determine whether or not a trailer is connected.

- DTC C2806 (Trailer Brakes Wiring Circuit Short to Vbatt) — this DTC sets in continuous memory when the TBC module detects that the battery voltage is greater than 9 volts while not braking. The TBC module shuts off the trailer brake output and then shuts down.
- DTC C2807 (Trailer Brakes Wiring Circuit Short to Ground) — this DTC sets in continuous memory or on-demand when the TBC module detects that the battery voltage drops by more than 20 percent during a braking event or the trailer brakes are drawing more than 28 amps. The TBC module shuts off the trailer brake output and then shuts down.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TBC module
- Customer trailer

PINPOINT TEST A : THE MESSAGE CENTER DISPLAYS WIRING FAULT ON TRAILER	
NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.	
NOTE: Make sure the trailer brakes are electric-magnet actuated drum type brakes before proceeding with this pinpoint test.	
A1 VERIFY TBC MODULE DTCS	
• Verify a trailer is not connected to the trailer tow connector. • Ignition ON. • Enter the following diagnostic mode on the scan tool: TBC Module Self-Test . • Clear the DTCs. • Test drive the vehicle in the following manner: ▪ maintain a speed greater than 5 km/h (3 mph) ▪ turn the headlights ON ▪ lightly apply the brakes for more than 5 seconds ▪ activate the left and right turn signals • Ignition OFF. • Ignition ON. • Wait at least 10 seconds for all modules to wake up and prove out.	

- Enter the following diagnostic mode on the scan tool: **TBC Module Self-Test** .

Is DTC C2806 or C2807 present?

Yes	GO to A2 .
No	For all other TBC module DTCs, GO to the TBC Module DTC Chart. If no DTCs return, the system is operating correctly at this time. INSPECT the 7-pin trailer tow connector for water intrusion, corrosion, pushed-out pins and spread terminals. REPAIR or INSTALL new as necessary. If the 7-pin trailer tow connector is OK, SEND the trailer to an authorized camper/trailer repair facility.

A2 CHECK THE TRAILER TOW CONNECTOR

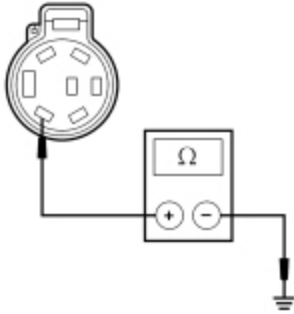
- Ignition OFF.
- Disconnect: Trailer Tow [C439A](#).
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **TBC Module Self-Test** .
- Clear the DTCs.
- Test drive the vehicle in the following manner:
 - maintain a speed greater than 5 km/h (3 mph)
 - turn the headlights ON
 - lightly apply the brakes for more than 5 seconds
 - activate the left and right turn signals
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds for all modules to wake up and prove out.
- Enter the following diagnostic mode on the scan tool: **TBC Module Self-Test** .

Is DTC C2806 or C2807 present?

Yes	For DTC C2806, GO to A4 . For DTC C2807, GO to A3 .
No	INSTALL a new 7-pin trailer tow connector. CLEAR any DTCs present. TEST the system for normal operation.

A3 CHECK THE TBC MODULE PWM OUTPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Connect: Trailer Tow [C439A](#).
- Disconnect: **TBC** Module [C2142](#).
- Measure the resistance between [C439A](#) Pin 3, circuit CAT19 (BU), harness side and ground.



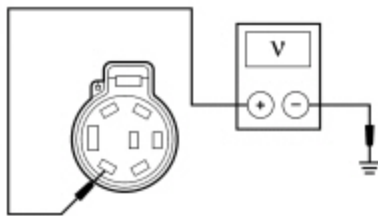
• N0009553

Is the resistance greater than 10,000 ohms?

Yes	GO to A5 .
No	REPAIR circuit CAT19 (BU) for a short to ground. TEST the system for normal operation.

A4 CHECK THE TBC MODULE PWM OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Trailer Tow [C439A](#).
- Disconnect: TBC Module [C2142](#).
- Ignition ON.
- Measure the voltage between [C439A](#) Pin 3, circuit CAT19, (BU), harness side and ground while operating the following lamps in turn:
 - Parking lamps
 - Stop lamps
 - Reversing lamps



• A0025430

Is any voltage present?

Yes	REPAIR circuit CAT19 (BU) for a short to power. TEST the system for normal operation.
No	GO to A5 .

A5 CHECK FOR CORRECT TBC MODULE OPERATION

- Ignition OFF.
- Inspect the TBC module connector.
- Repair:

- corrosion (install a new connector or terminal - clean module pins)
 - damaged or bent pins - install new terminals or pins
 - pushed-out pins - install new pins as necessary
 - spread terminals - install new terminals as necessary
- Connect the **TBC** module connector and make sure that it seats correctly.
 - Test drive the vehicle in the following manner:
 - maintain a speed greater than 5 km/h (3 mph)
 - turn the headlights ON
 - lightly apply the brakes for more than 5 seconds
 - activate the left and right turn signals
 - Ignition OFF.
 - Ignition ON.
 - Wait at least 10 seconds for all modules to wake up and prove out.
 - Enter the following diagnostic mode on the scan tool: **TBC** Module Self-Test .

Is DTC C2806 or C2807 present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: The Trailer Brakes are Inoperative

NOTE: Carry out all pinpoint tests *without* a trailer connected to the vehicle.

Refer to Wiring Diagrams Cell [95](#) , Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The braking energy provided to the trailer is varied with a Pulse Width Modulated (PWM) signal that switches between 0 volts and battery voltage, the higher the duty cycle the more braking power available. The Trailer Brake Control (TBC) module varies the PWM signal based on various inputs such as the brake torque message, the manual slider switch and the gain buttons. With the vehicle stationary and the manual slider switch fully to the left with a gain of 10, there should be more than 10-12 volts supplied to the trailer tow connector. The **TBC** module Pulse Width Modulated (PWM) output voltage is speed dependent. With the vehicle stationary and the brake pedal applied, there is approximately 0 volts at the trailer tow connector (pin-3 or pin-7).

The **TBC** module receives stoplamp switch information from the PCM and information from the ABS module over the High Speed Controller Area Network (HS-CAN) . If there is a problem in the ABS module that is preventing this information from being sent to the **TBC** module, the **TBC** module enters a limited operating mode. The **TBC** module receives fused battery voltage from Battery Junction Box (BJB) fuse 17 (30A) and fused ignition voltage from Body Control Module (BCM) fuse 37 (10A).

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors

- Trailer tow connector
- TBC module
- Customer trailer

PINPOINT TEST B : THE TRAILER BRAKES ARE INOPERATIVE

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: Make sure the trailer brakes are electric-magnet actuated drum-type brakes before proceeding with this pinpoint test.

B1 CHECK FOR ANY PCM DTCS

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Enter the following diagnostic mode on the scan tool: Self Test — PCM Key ON Engine OFF (KOEO) .
- Retrieve and record any PCM DTCS.

Are any PCM DTCS retrieved?

Yes	REFER to Section 303-14.
No	GO to B2 .

B2 CHECK FOR ANY ABS MODULE DTCS

- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Retrieve and record any ABS module DTCS.

Are any ABS module DTCS retrieved?

Yes	REFER to Section 206-09.
No	GO to B3 .

B3 CHECK FOR SCAN TOOL COMMUNICATION WITH THE TBC MODULE

- Using a scan tool, carry out the network test.

Did the TBC module pass the network test?

Yes	GO to B4 .
No	REFER to Section 418-00.

B4 CHECK FOR TBC MODULE DTCS

- Review the DTCS retrieved and recorded during TBC module self-test.

Are any TBC module DTCS present?

Yes	REFER to the Trailer Brake Control (TBC) Module DTC Chart.
No	GO to B5 .

B5 CHECK THE TBC MODULE FOR MAXIMUM OUTPUT

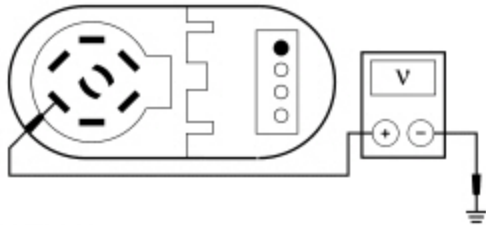
NOTE: Battery voltage must be greater than 10 volts.

- Ignition ON.
- Set a gain of 10 in the TBC module.
- **NOTE:** If the manual slider switch is held to the full left position for more than 5 to 20 minutes, the TBC module automatically shuts down.

Have an assistant press the manual slider switch fully to the left with a gain of 10 on the TBC module.

- **NOTE:** Do not apply the brake pedal during this test.

Measure the voltage between trailer tow 7-pin trailer tow connector electric brake supply pin, component side and ground.



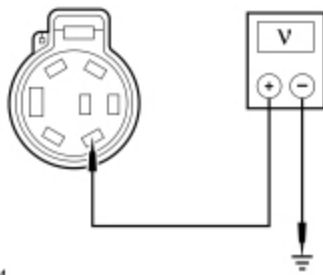
• N0084023

Is the voltage greater than 10-12 volts?

Yes	The TBC module is operating correctly. SEND the trailer to an authorized camper/trailer repair center for an overload condition.
No	GO to B6 .

B6 CHECK THE TRAILER TOW CONNECTOR FOR AN OPEN

- Ignition OFF.
- Disconnect: Trailer Tow Connector [C439A](#).
- Ignition ON.
- Have an assistant press the manual slider switch fully to the left with a gain of 10 on the TBC module.
- Measure the voltage between the trailer tow connector [C439A](#) Pin 3, circuit CAT19 (BU), harness side and ground.



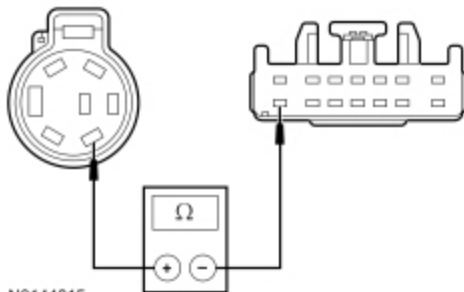
N0144814

Is the voltage greater than 10 volts?

Yes	INSTALL a new trailer tow connector. CLEAR the DTCs. TEST the system for normal operation.
No	GO to B7 .

B7 CHECK THE TBC MODULE PWM OUTPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the trailer tow connector [C439A](#) Pin 3, circuit CAT19 (BU), harness side and TBC module [C2142](#) Pin 14, circuit CAT19 (BU), harness side.



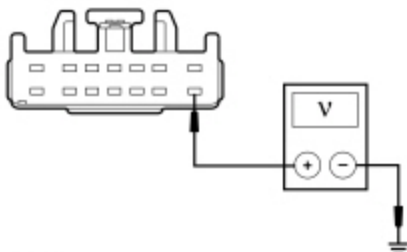
N0144815

Is the resistance less than 5 ohms?

Yes	GO to B8 .
No	REPAIR the circuit. TEST the system for normal operation.

B8 CHECK THE BATTERY VOLTAGE TO THE TBC MODULE

- Ignition OFF.
- Measure the voltage between TBC module [C2142](#) Pin 8, circuit SBB17 (RD), harness side and ground.



N0121967

Is the voltage greater than 10 volts?

Yes	GO to B9 .
No	VERIFY the BJB fuse 17 (30A) is OK. If OK, REPAIR circuit SBB17 (RD) for an open. TEST the system for normal operation.

B9 CHECK FOR CORRECT TBC MODULE OPERATION

- Disconnect the TBC module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the TBC module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: The Trailer Brakes are Always Engaged

NOTE: Carry out all pinpoint tests *without* a trailer connected to the vehicle.

Refer to Wiring Diagrams Cell [95](#) , Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The braking energy provided to the trailer is varied with a Pulse Width Modulated (PWM) signal that switches between 0 volts and battery voltage, the higher the duty cycle the more braking power available. With the vehicle stationary and the manual slider switch fully to the left with a gain of 10, there should be more than 10-12 volts supplied to pin 3 of the 7-pin trailer tow connector. The TBC module Pulse Width Modulated (PWM) output voltage is speed dependent. With the vehicle stationary and the brake pedal applied, there is approximately 0 volts at the trailer tow connector (pin-3 or pin-7). The Trailer Brake Control (TBC) module also receives information from the brake pressure switch (also called the speed control deactivation switch) and the stoplamp switch from the PCM over the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TBC module
- Customer trailer

PINPOINT TEST C : THE TRAILER BRAKES ARE ALWAYS ENGAGED

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

C1 CHECK FOR ANY PCM DTCS

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Enter the following diagnostic mode on the scan tool: Self Test — PCM Key ON Engine OFF (KOEO) .
- Retrieve and record any PCM DTCS.

Are any PCM DTCS retrieved?

Yes	REFER to Section 303-14.
No	GO to C2 .

C2 CHECK FOR ANY TBC MODULE DTCS

- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .
- Retrieve and record any TBC module DTCS.

Are any TBC module DTCS retrieved?

Yes	REFER to DTC Charts in this section.
No	GO to C3 .

C3 OBSERVE THE VEHICLE STOPLAMP OPERATION

- Monitor the vehicle stoplamps while pressing and releasing the brake pedal.

Do the vehicle stoplamps operate correctly?

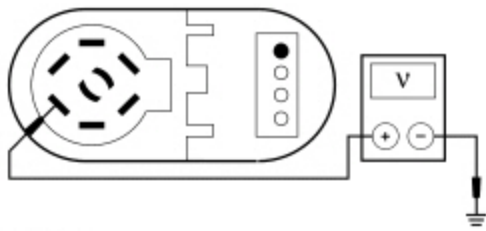
Yes	GO to C4 .
No	REFER to Section 417-01 to diagnose the stoplamp switch.

C4 CHECK THE TBC MODULE OUTPUT

NOTE: Every 4 seconds a voltage pulse is applied as a trailer connectivity test.

- Ignition ON.
- Make sure the brake pedal is not pressed and the manual slider switch is not being applied.
- **NOTE:** Do not apply the brake pedal during this test.

Measure the voltage between 7-pin trailer tow connector electric brake supply pin, component side and ground.



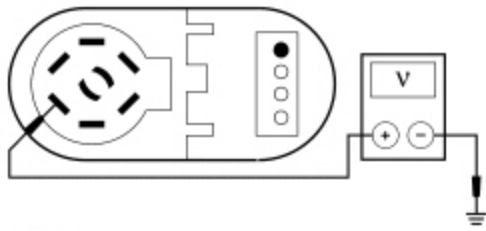
• N0084023

Is the voltage approximately 0.3-0.4 volt?

Yes	The TBC module and vehicle wiring are operating correctly. SEND the trailer to an authorized camper/trailer repair center.
No	GO to C5 .

C5 CHECK THE TBC MODULE PWM OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: TBC Module [C2142](#).
- Ignition ON.
- Measure the voltage between 7-pin trailer tow connector electric brake supply pin, component side and ground.



• N0084023

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to C6 .

C6 CHECK FOR CORRECT TBC MODULE OPERATION

- Disconnect the TBC module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the TBC module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test D: DTC B1317

Refer to Wiring Diagrams Cell 13 , Power Distribution/BCM for schematic and connector information.

Normal Operation

- DTC B1317 (Battery Voltage High) — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module when the TBC module detects that battery voltage has risen above 16 volts on the voltage supply circuit while not braking.

This pinpoint test is intended to diagnose the following:

- Charging system concern
- TBC module

PINPOINT TEST D : DTC B1317

NOTE: DTC B1317 may be stored in the module memory due to previous battery charging or vehicle jump starting events.

D1 CHECK FOR DTC B1317, B1676, P0563 (PCM) OR U3003:17 SET IN OTHER MODULES

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Retrieve the continuous memory DTCs from all modules.

Is DTC B1317, B1676, P0563 (PCM) or U3003:17 set in more than one module?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to D2.

D2 CHECK THE BATTERY VOLTAGE

- Turn off all interior/exterior lights and accessories.
- Start and run the engine at approximately 2,000 rpm for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 16 volts or higher?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to D3.

D3 RECHECK FOR DTC B1317

- Turn the engine off.
- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .
- Clear the continuous memory DTCs.
- Carry out the TBC module self-test.

Is DTC B1317 present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module module in this section. TEST the system for normal operation.
No	The system is operating normally at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle.

Pinpoint Test E: DTC B1318

Refer to Wiring Diagrams Cell 13 , Power Distribution/SJB for schematic and connector information.

Normal Operation

- DTC B1318 (Battery Voltage Low) — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module when the TBC module detects that battery voltage has dropped below 9 volts on the voltage supply circuit while not braking.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Charging system concern
- TBC module

PINPOINT TEST E : DTC B1318

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.

E1 RECHECK THE TBC MODULE DTCS

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .
- Clear the DTCs. Repeat the TBC module self-test.

Is DTC B1318 still present?

Yes	GO to E2 .
No	The system is operating normally at this time. The DTC may have been set due to a previous low battery voltage condition.

E2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM

- Enter the following diagnostic mode on the scan tool: Self Test — PCM Key ON Engine OFF (KOEO) .

Is DTC P0620, P0622, P0625, P0626 or P065B set in the PCM?

Yes	REFER to Section 414-01.
No	GO to E3 .

E3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE

- Check the battery condition and verify the battery is fully charged. Refer to Section 414-01.

Is the battery OK and fully charged?

Yes	GO to E4 .
No	REFER to Section 414-00.

E4 CHECK THE TBC MODULE VOLTAGE (MODULE_VOLT) PID

- Ignition ON.
- Measure and record the voltage at the battery.
- Enter the following diagnostic mode on the scan tool: DataLogger — TBC Module .
- Monitor the TBC MODULE_VOLT PID.

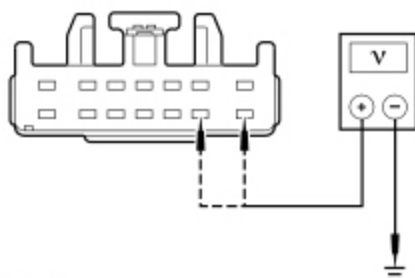
Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to E7 .
No	GO to E5 .

E5 CHECK THE TBC MODULE VOLTAGE SUPPLY

- Ignition OFF.
- Disconnect: TBC Module [C2142](#).
- Ignition ON.

- Measure the voltage between the TBC module [C2142](#) Pin 8, circuit SBB17 (RD), harness side and ground; and between the TBC module [C2142](#) Pin 9, circuit CBP37 (WH), harness side and ground.



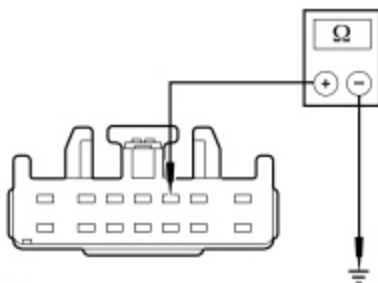
N0091204

Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to E6 .
No	REPAIR the circuit. TEST the system for normal operation.

E6 CHECK THE TBC MODULE GROUND CIRCUIT FOR HIGH RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the TBC module [C2142](#) Pin 3, circuit GD138 (BK/WH), harness side and ground.



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Is the resistance less than 5 ohms?

Yes	GO to E7 .
No	REPAIR the circuit. CLEAR the DTC. REPEAT the self-test.

E7 CHECK FOR CORRECT TBC MODULE OPERATION

- Disconnect the TBC module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the TBC module connector and make sure it seats correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: DTC B1863

Refer to Wiring Diagrams Cell [95](#) , Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The Trailer Brake Control (TBC) module receives battery voltage from Battery Junction Box (BJB) fuse 17 (30A) and ignition voltage from Body Control Module (BCM) fuse 37 (10A).

- DTC B1863 (TBC Ground Open Circuit) — this DTC sets in continuous memory or on-demand when the TBC module detects an open circuit in the wiring, a connector condition causes an open in the ground circuit or if an internal failure of the TBC module causes an open in the ground circuit. The TBC module also disables trailer braking and sends a TRAILER BRAKE MODULE FAULT message to the message center. A trailer must be connected for this DTC to set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TBC module

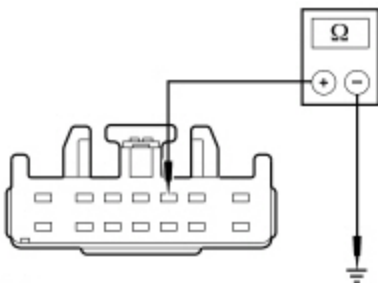
PINPOINT TEST F : DTC B1863

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: A trailer must be connected to the 7-pin trailer connector for DTC B1863 to set during the on-demand self-test.

F1 CHECK THE TBC MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: TBC Module [C2142](#).
- Measure the resistance between TBC module [C2142](#) Pin 3, circuit GD138 (BK/WH), harness side and ground.



Is the resistance less than 5 ohms?

Yes	GO to F2 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

F2 CHECK FOR CORRECT TBC MODULE OPERATION

- Disconnect the TBC module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
 - broken harness wire
 - broken connector
- Connect the TBC module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test G: DTC U0100

Normal Operation

- DTC U0100 (Lost Communication With ECM/PCM "B") — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module if data messages received (vehicle speed) from the PCM are missing for more than 5 seconds. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- PCM
- TBC module

PINPOINT TEST G : DTC U0100

G1 VERIFY THE SCAN TOOL CAN COMMUNICATE WITH THE PCM

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Verify that a vehicle session can be established with a scan tool.

Can a vehicle session be established?

Yes	GO to G2 .
No	REFER to Section 418-00 to diagnose no communication with the PCM.

G2 CHECK THE TBC MODULE CONTINUOUS MEMORY DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .
- Clear the DTCs. Repeat the TBC module self-test. Record all TBC module DTCs.

Is DTC U0100 retrieved again?

Yes	GO to G3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

G3 RETRIEVE THE RECORDED DTCS FROM THE TBC MODULE SELF-TEST

- Check for recorded TBC module DTCs from the previous self-test.

Is DTC B1317 or DTC B1318 recorded?

Yes	For DTC B1317, GO to Pinpoint Test D . For DTC B1318, GO to Pinpoint Test E .
No	GO to G4 .

G4 RETRIEVE THE RECORDED DTCS FROM THE PCM SELF-TEST

- Enter the following diagnostic mode on the scan tool: Self Test — PCM Key ON Engine OFF (KOEO) .

Is DTC P0562 or DTC P0563 recorded?

Yes	REFER to Section 303-14.
No	GO to G5 .

G5 CHECK FOR DTC U0100:00 SET IN OTHER MODULES

- Using a scan tool, retrieve the continuous memory DTCs from the following modules:
 - Accessory Protocol Interface Module (APIM) (if equipped)
 - BCM
 - Instrument Panel Cluster (IPC)
 - Occupant Classification System Module (OCSM)
 - Power Steering Control Module (PSCM) (if equipped)
 - Restraints Control Module (RCM)
 - Transfer Case Control Module (TCCM) (if equipped)

Is DTC U0100:00 set in any of these modules?

Yes	INSTALL a new PCM. REFER to Section 303-14.
No	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.

Pinpoint Test H: DTC U0121

Normal Operation

- DTC U0121 (Lost Communication With Anti-Lock Brake System (ABS) Control Module) — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module if data messages such as brake torque information are not received from the ABS module. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ABS module
- TBC module

PINPOINT TEST H : DTC U0121

H1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to H2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

H2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out the network test.

Does the ABS module pass the network test?

Yes	GO to H3 .
No	REFER to Section 418-00 to diagnose the ABS module does not respond to the scan tool.

H3 RETRIEVE ANY DTCS FROM THE TBC MODULE

- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .

Is DTC U3003:16 (or B1318) or DTC U3003:17 (or B1317) recorded?

Yes	For DTC B1317, GO to Pinpoint Test D . For DTC B1318, GO to Pinpoint Test E . For the ABS module, REFER to Section 206-09.
No	GO to H4 .

H4 CHECK FOR DTC U0121:00 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Wait at least 5 seconds to allow the ~~TBC~~ module to sleep (power down).
- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Using a scan tool, retrieve the continuous memory DTCs from the following modules:
 - Body Control Module (BCM)
 - Instrument Panel Cluster (IPC)
 - Power Steering Control Module (PSCM) (if equipped)
 - Transfer Case Control Module (TCCM) (if equipped)

Is DTC U0121:00 set in any of these modules?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CLEAR the DTCs. REPEAT the ABS module self-test.
No	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.

Pinpoint Test I: DTC U0140

Normal Operation

- DTC U0140 (Lost Communication With Body Control Module (GEM)) — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module if data messages such as ignition state, park lamp status and illumination dimming level data are not received from the Body Control Module (BCM) . For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ~~BCM~~
- ~~TBC~~ module

PINPOINT TEST I : DTC U0140

I1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.

- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to I2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

I2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out the network test.

Does the BCM module pass the network test?

Yes	GO to I3 .
No	REFER to Section 418-00.

I3 RETRIEVE ANY DTCS FROM THE TBC MODULE

- Enter the following diagnostic mode on the scan tool: Self Test — ~~TBC~~ Module .

Is DTC B1317 or DTC B1318 recorded?

Yes	For DTC B1317, GO to Pinpoint Test D . For DTC B1318, GO to Pinpoint Test E .
No	GO to I4 .

I4 RETRIEVE ANY DTCS FROM THE BCM

- Enter the following diagnostic mode on the scan tool: Self Test — ~~BCM~~ .

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 419-10.
No	GO to I5 .

I5 RECHECK THE TBC MODULE DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs. Repeat the ~~TBC~~ module self-test.

Is DTC U0140 still present?

Yes	GO to I6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

I6 CHECK FOR DTC U0140 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Wait at least 5 seconds to allow the **TBC** module to sleep (power down).
- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Using a scan tool, retrieve the continuous memory DTCs from all modules.

Is DTC U0140 set in more than one module?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . REPEAT the BCM self-test.
No	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.

Pinpoint Test J: DTC U0155

Normal Operation

- DTC U0155 (Lost Communication With Instrument Panel Cluster (IC) Control Module) — this DTC sets in continuous memory or on-demand in the Trailer Brake Control (TBC) module if data messages received from the Instrument Panel Cluster (IPC) are missing. The **IPC** sends a TRAILER BRAKE CONFIG message to the **TBC** module. For a complete list of all network messages, refer to Section 413-01.

This pinpoint test is intended to diagnose the following:

- **IPC**
- **TBC** module

PINPOINT TEST J : DTC U0155

J1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to J2 .

No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
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J2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out the network test.

Does the IPC pass the network test?

Yes	GO to J3 .
No	REFER to Section 418-00.

J3 RETRIEVE THE RECORDED DTCS FROM THE TBC MODULE AND IPC SELF-TESTS

- Enter the following diagnostic mode on the scan tool: Self Test — TBC Module .
- Record the TBC module DTCs.
- Enter the following diagnostic mode on the scan tool: Self Test — IPC .

Is DTC U3003:16 (or B1318) or DTC U3003:17 (or B1317) recorded?

Yes	For TBC module DTC B1317, GO to Pinpoint Test D . For TBC module DTC B1318, GO to Pinpoint Test E . For IPC DTCs, REFER to Section 413-01.
No	GO to J4 .

J4 CHECK FOR DTC U0155 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Wait at least 5 seconds to allow the TBC module to sleep (power down).
- Ignition ON.
- Allow at least 10 seconds for all modules to wake up and prove out.
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0155:00 or DTC U0155 set in more than one module?

Yes	INSTALL a new IPC . REFER to Section 413-01. CLEAR the DTCs. REPEAT the IPC self-test.
No	INSTALL a new TBC module. REFER to Trailer Brake Control (TBC) Module in this section. TEST the system for normal operation.